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AIM:
To find the orders per customer, the average quantity per product and the earliest and latest order dates.

Software Used:
Python 3, Pandas (Jupyter Notebook)

Algorithm:
1. Create the Dataframe order_data and convert the data column to determine.
2. Group by Customer ID and count the orders.
3. Group by product & take the mean quantity.
4. Find the minimum and maximum dates.
5. Display the results.

CODE:

```
import panda as pd
order_data = pd.DataFrame({
        "customer_id": [101, 102, 101, 103, 102, 101, 104, 103
        "order_date": ["2024-01-05", "2024-01-12", "2024-02-03
                       "2024-02-15", "2024-03-01", "2024-03-25
        "product_name": ["Laptop", "mouse", "mouse",
                         "Keyboard"],
        "order_quantity": [1, 3, 2, 4, 2, 1, 5, 3]
```

3)

```
orders_data["order_date"] = pd.to_datetime(orders_data["order_date"])
print(orders_data.groupby("customer_id").size())
print(orders_data.groupby("product_name")["order_quantity"].mean())
print(orders_data["order_date"].min().date(), orders_data["order_date"].max().date())
```

Input/Output:

Customer_id: 101 → 3, 102 → 2, 103 → 2, 104 → 1

Keyboard 2.50 | Laptop 1.50 | Mouse 3.25

2024-01-05 2024-04-02.

Result:

Customer 101 placed the most orders (3). Mouse has the highest average quantity (3.25).

Order span 05-Jan-2024 to 02-Apr-2024.